Whatever the eleven-dimensional theory is, Witten has provisionally named it M-theory. The name stands for as many things as people you poll. Some samples: Mystery Theory, Mother Theory (as in “Mother of all Theories”), Membrane Theory (since, whatever it is, membranes seem to be part of the story), Matrix Theory (after some recent work by Tom Banks of Rutgers University, Willy Fischler of the University of Texas at Austin, Stephen Shenker of Rutgers University, and Susskind that offers a novel interpretation of the theory). But even without having a firm grasp on its name or its properties, it is already clear that M-theory provides a unifying substrate for pulling together all five string theories.

M-Theory and the Web of Interconnections

There is an old proverb about three blind men and an elephant. The first blind man grabs hold of the elephant’s ivory tusk and describes the smooth, hard surface that he feels. The second blind man grabs hold of one of the elephant’s legs. He describes the tough, muscular girth that he feels. The third blind man grabs hold of the elephant’s tail and describes the slender and sinewy appendage that he feels. Since their mutual descriptions are so different, and since none of the men can see the others, each thinks that he has grabbed hold of a different animal. For many years, physicists were as much in the dark as the blind men, thinking that the different string theories were very different. But now, through the insights of the second superstring revolution, physicists have realized that M-theory is the unifying pachyderm of the five string theories.

In this chapter we have discussed changes in our understanding of string theory that arise when we venture beyond the domain of the perturbative framework—a framework implicitly in use prior to this chapter. Figure 12.9 summarizes the interrelations we have found so far, with arrows to indicate dual theories. As you can see, we have a web of connections, but it is not yet complete. By also including the dualities of Chapter 10, we can finish the job.

Recall the large/small circular radius duality that interchanges a circular dimension of radius R with one whose radius is 1/R. Previously, we glossed over one aspect of this duality, which we now must clarify. In Chapter 10, we discussed the properties of strings in a universe with a circular dimension without carefully specifying which